

Candidate Pseudolabel Learning: Enhancing VLMs with Unlabeled Data

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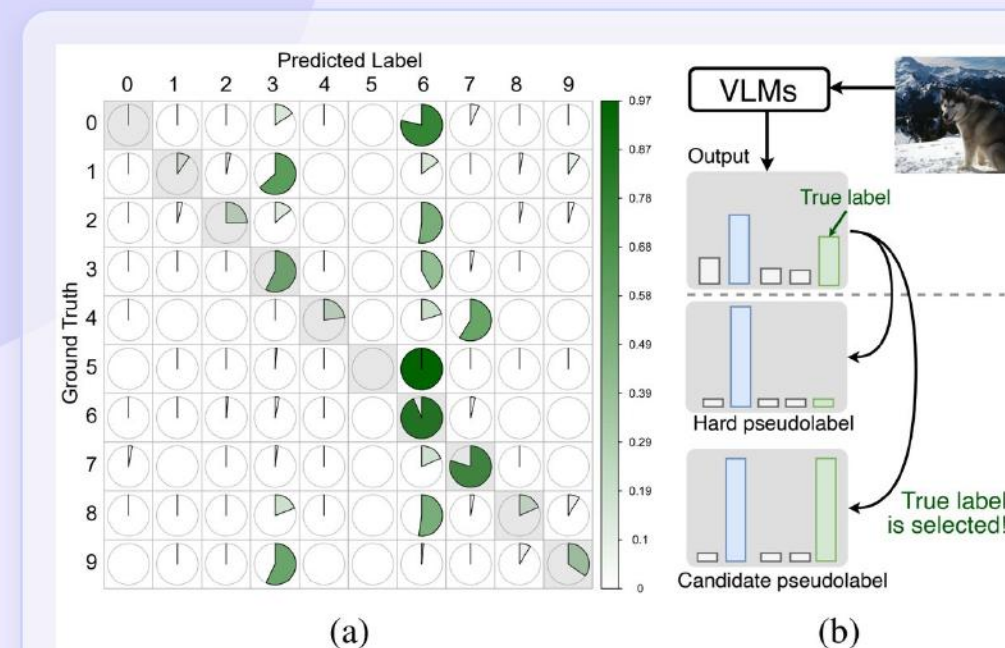
01 Zero-shot VLMs and Labeling Bottleneck

- **Promise of zero-shot VLMs**, yet adaptation still needs labels
 - Labeled data is costly across diverse tasks and domains
- Unlabeled data is plentiful; pseudolabeling is attractive
 - **Hard pseudolabels are often inaccurate and class-imbalanced**
- Goal: adapt VLMs with minimal labels via **more reliable candidate pseudolabels**
 - Reduce reliance on high zero-shot accuracy and manual annotation

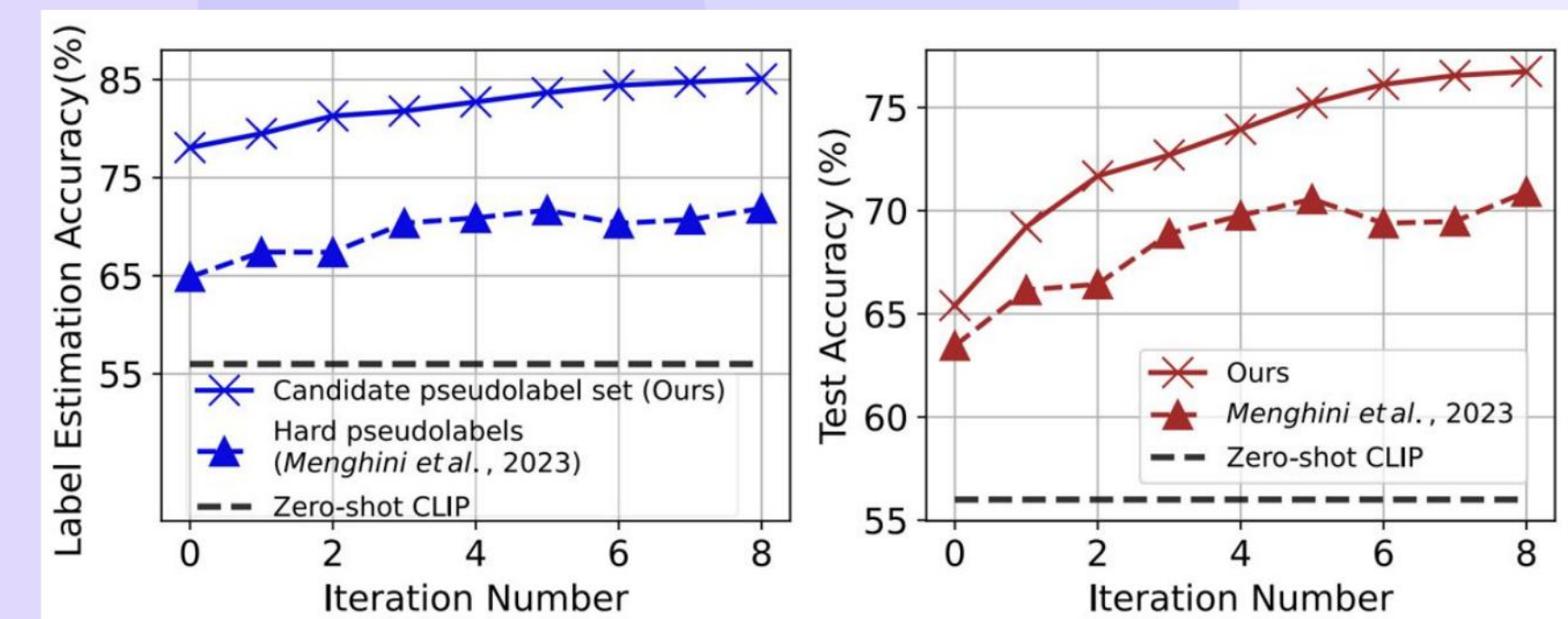
Radford et al., 2021; Li et al., 2022; Yuan et al., 2021; Zhou et al., 2022a; Zhou et al., 2022b; Zhang et al., 2024

02 Hard Pseudolabel Pitfalls (EuroSAT)

- Hard pseudolabels skew toward dominant classes
 - EuroSAT shows many samples misassigned to a single class
- **Fine-tuning on wrong and imbalanced labels degrades accuracy**
 - Systematic errors propagate during training without correction
- Confusion analysis reveals **consistent bias across classes**
 - Motivates moving beyond top-one labels



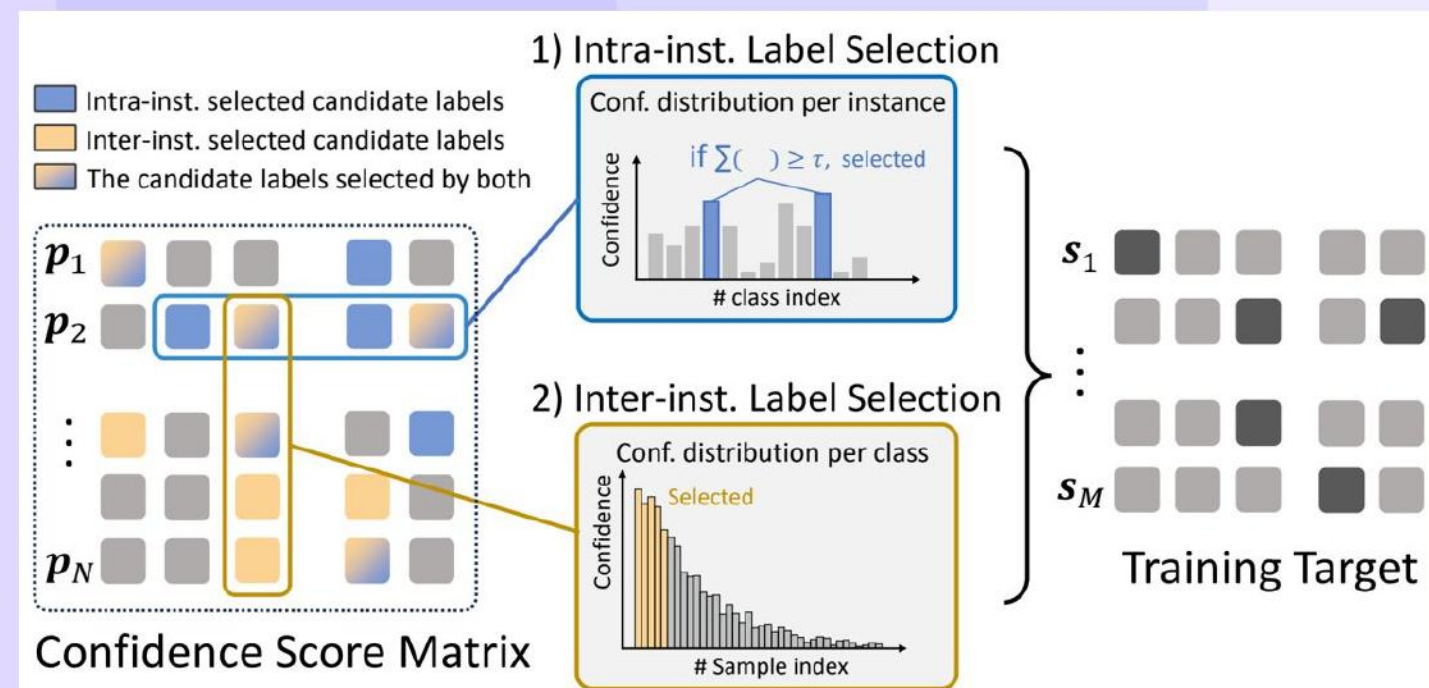
03 Candidate Sets: Intuition and Empirical Edge



- Use a set of top candidate labels per instance
 - Higher chance the true label is included than top-one
- Empirically improves label estimation and test accuracy
 - Performance rises across iterations on multiple datasets
- Treat all candidates equally to dilute class prior bias
 - Aligns with principles from partial-label learning

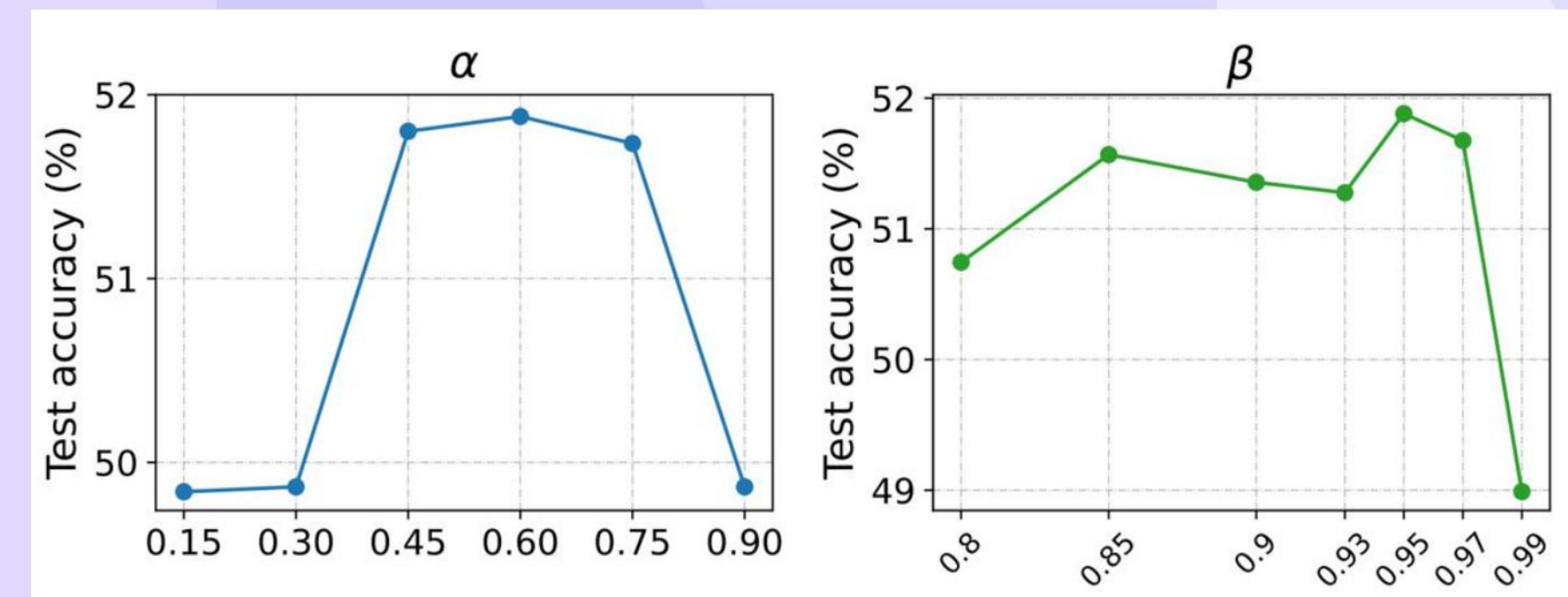
Helber et al., 2019; Radford et al., 2021; Menghini et al., 2023

Hossain and Kauranen, 2015; Li et al., 2023; Menghini et al., 2023

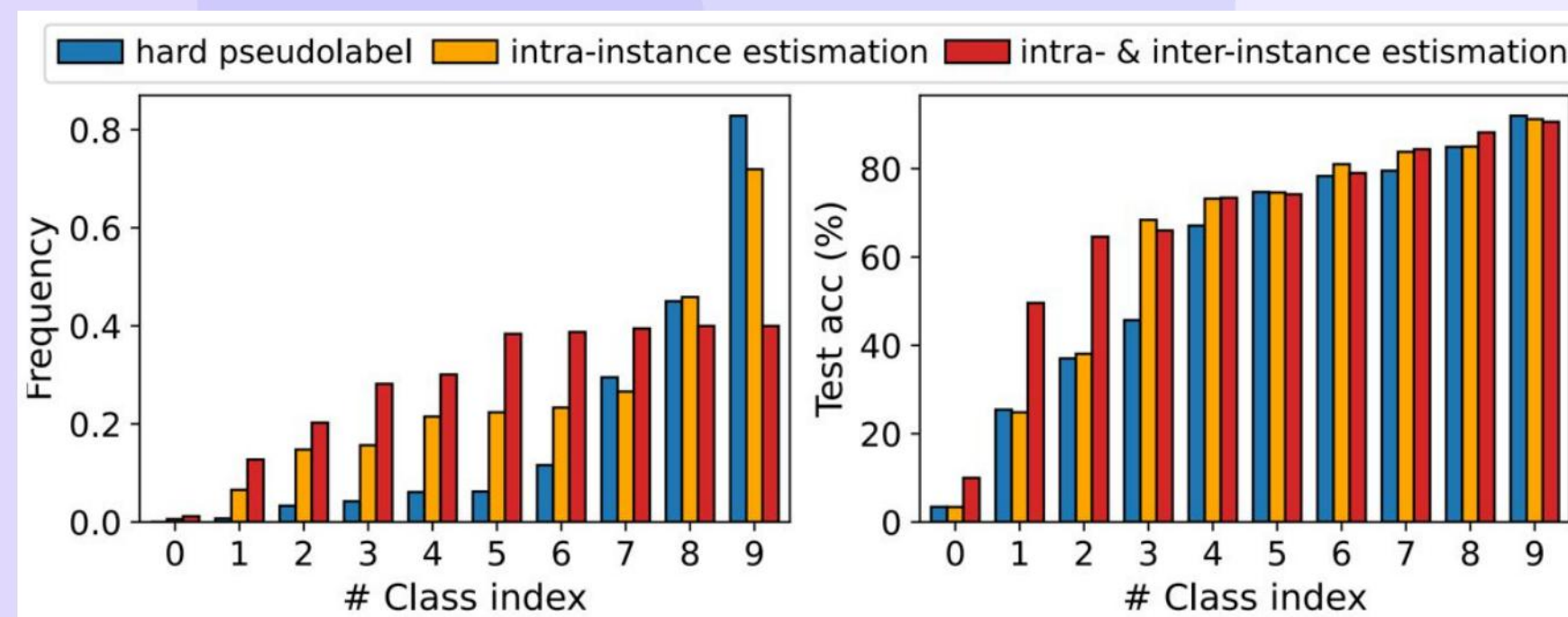


- Construct a confidence matrix over all unlabeled data
 - Each row stores per-class scores for one instance
- Generate candidate sets and train with partial-label losses
 - Iteratively refine candidates and the model parameters
- Balance classes via inter-instance selection and adapt the confidence threshold by a chosen quantile
 - Convert multiclass supervision into candidate-label supervision

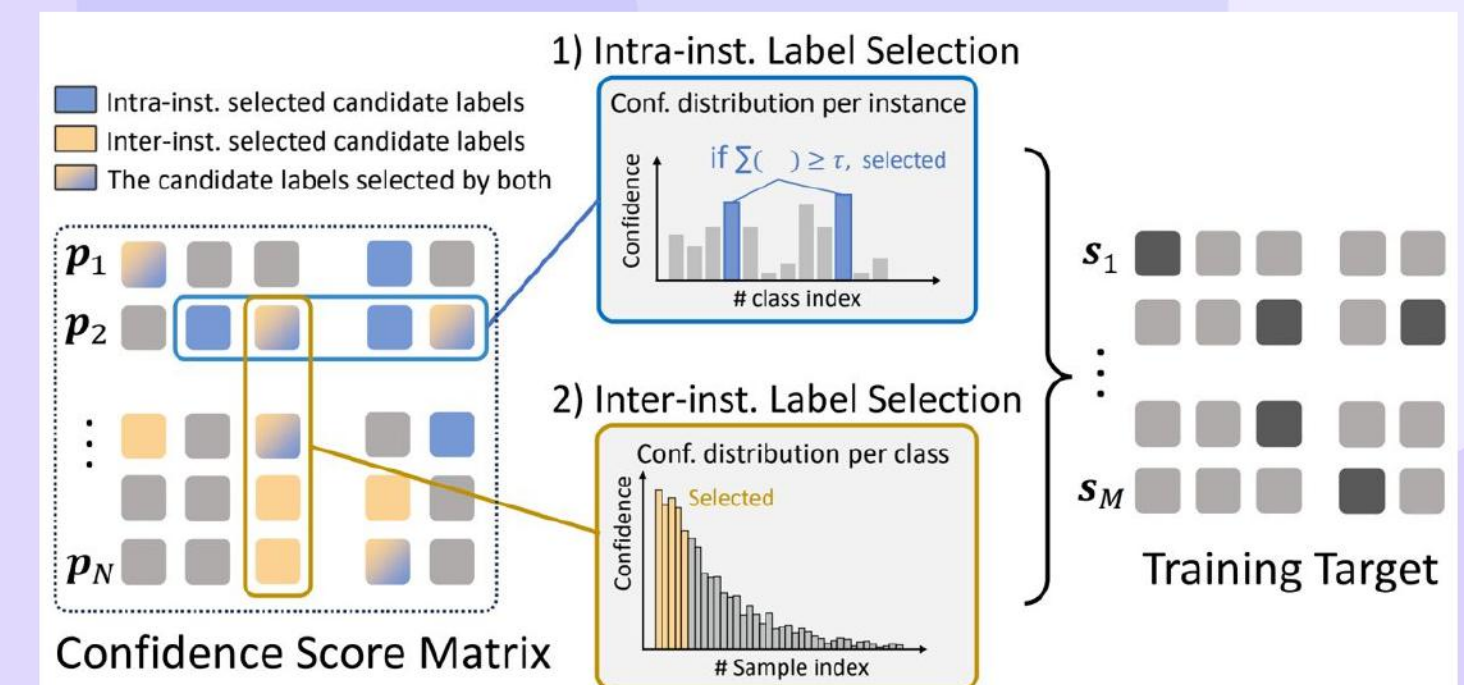
Cour et al., 2011; Luo and Orabona, 2010



- Accumulate sorted class confidences until the sum reaches a threshold
 - Instance-specific candidate set size reflects difficulty
- The threshold adapts each iteration using a specified quantile of maximum confidences
 - Tracks improving calibration during training
- Limits ambiguity by enforcing comparable total confidence per instance
 - Reduces overly large candidate sets that hinder learning



- Per-class selection keeps top confident instances
 - Mitigates dominance of frequent or easy classes
- Final candidate set equals the intersection of intra- and inter-instance selections
 - Retains confident and balanced label options
- Tune the selection ratio to avoid empty sets and maintain coverage
 - Stabilizes iterative updates and class balance



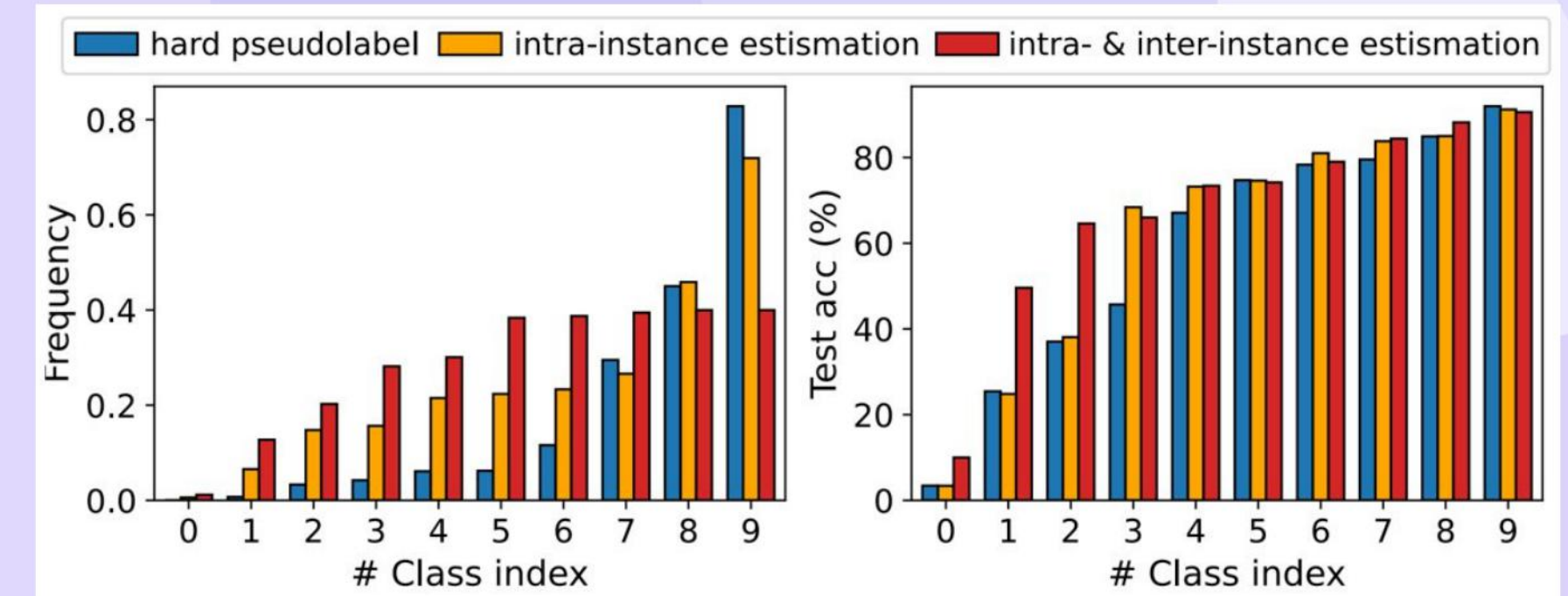
- Adopt standard partial-label losses such as CC, RC, CAV, and LW
 - Flexible and effective across tasks
- Iterate: regenerate candidates and continue fine-tuning
 - Supports semi-supervised, unsupervised, and transductive zero-shot settings
- Construct binary target vectors from candidate sets and filter empty-candidate instances
 - Simple integration into existing training loops

Feng et al., 2020; Wen et al., 2021; Zhang et al., 2021b

Methods	Flowers102			RESISC45			DTD		
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL
Zero-shot CLIP	63.67 _{0.00}			54.48 _{0.00}			43.24 _{0.00}		
FPL	75.96 _{0.74}	65.67 _{0.23}	80.97 _{0.00}	68.13 _{0.55}	63.07 _{0.38}	72.11 _{0.00}	37.10 _{5.45}	44.96 _{0.55}	46.30 _{0.03}
CPL (Ours)	77.36 _{0.24}	70.01 _{0.21}	84.60 _{0.10}	71.73 _{0.57}	68.47 _{0.34}	72.16 _{0.26}	54.63 _{0.79}	48.92 _{0.17}	59.79 _{1.32}
GRIP	83.60 _{0.48}	69.84 _{1.06}	86.26 _{0.00}	74.11 _{0.68}	70.55 _{0.88}	81.07 _{0.00}	56.07 _{0.85}	46.09 _{1.06}	65.30 _{0.01}
CPL (Ours)	89.66 _{0.36}	72.90 _{0.78}	87.35 _{0.76}	80.98 _{0.11}	77.39 _{0.44}	85.85 _{0.49}	61.21 _{0.56}	51.91 _{0.71}	68.00 _{0.34}

Methods	CUB			EuroSAT			FGVCAircraft		
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL
Zero-shot CLIP	51.82 _{0.00}			32.88 _{0.00}			17.58 _{0.00}		
FPL	55.29 _{0.59}	53.04 _{0.53}	55.44 _{0.20}	62.05 _{1.64}	48.96 _{1.49}	53.70 _{26.87}	20.02 _{0.77}	16.62 _{0.67}	17.55 _{0.37}
CPL (Ours)	56.37 _{0.45}	54.18 _{0.05}	64.01 _{0.17}	64.84 _{2.15}	51.45 _{1.97}	54.03 _{2.27}	22.37 _{0.66}	18.90 _{0.20}	28.47 _{0.43}
GRIP	56.65 _{0.33}	51.42 _{0.21}	59.48 _{0.38}	58.66 _{2.64}	57.21 _{1.77}	92.33 _{0.69}	16.98 _{0.82}	15.22 _{0.71}	26.08 _{0.25}
CPL (Ours)	58.53 _{0.24}	53.47 _{0.36}	66.20 _{0.50}	77.51 _{0.80}	67.26 _{0.47}	93.78 _{0.12}	22.48 _{0.63}	18.35 _{0.27}	30.86 _{0.70}

- **CPL outperforms** hard pseudolabel baselines across datasets
 - Consistent gains with both textual and visual prompts
- Stronger improvements when zero-shot accuracy is weak
 - Notable on EuroSAT and DTD
- Works under semi-supervised, unsupervised, and transductive zero-shot paradigms
 - Scales with more unlabeled data



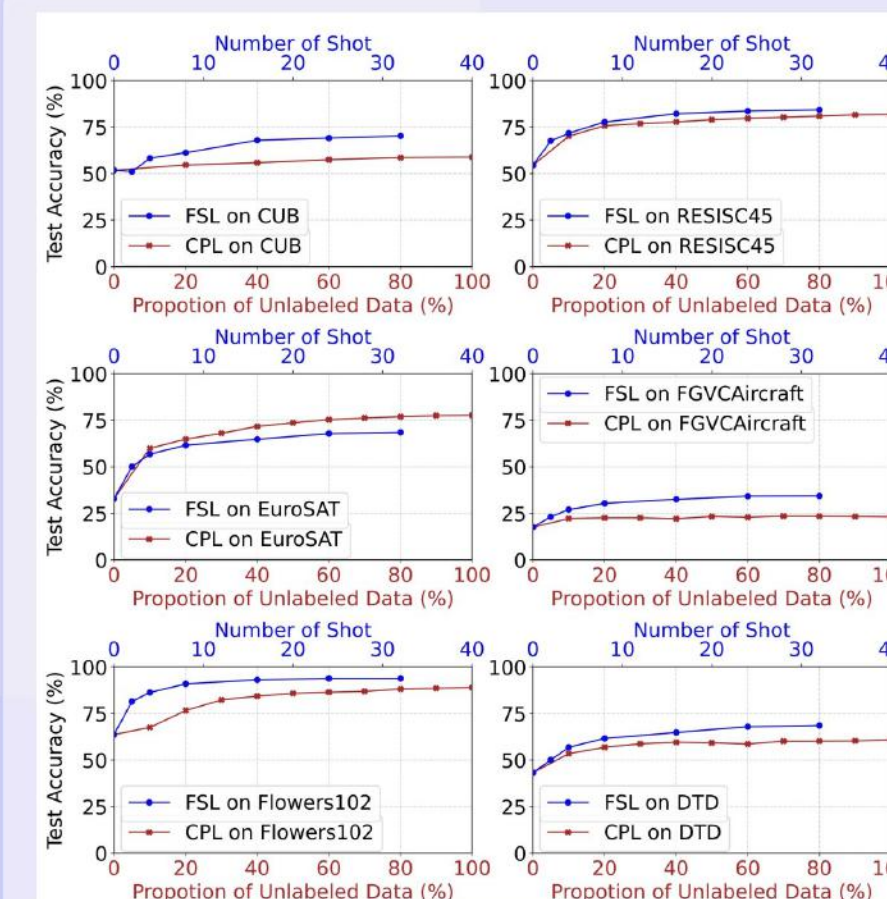
- **Less dependent** on initial zero-shot accuracy
 - More resilient when initial predictions are weak
- Combined selections yield **more balanced training targets**
 - Reduces skew from dominant classes
- Equal-weight candidates avoid amplifying class priors
 - Iterative updates progressively denoise labels

Huang et al., 2022; Menghini et al., 2023; Mirza et al., 2023

	Flowers-102	UCF-101	CIFAR-100	EuroSAT	DTD	CALTECH-101
CLIP	66.6	61.0	64.2	45.1	42.9	90.5
CLIP-PR	57.7	57.9	63.2	44.2	40.1	84.8
UPL	<u>71.5</u>	63.9	65.8	62.2	<u>48.0</u>	90.6
LaFTer	71.0	<u>68.2</u>	<u>74.6</u>	<u>73.9</u>	<u>46.1</u>	<u>93.3</u>
LaFTer + Ours	76.7	71.0	77.3	82.2	56.3	93.4

- CPL performs well with diverse partial-label losses
 - Competitive across CC, RC, CAV, LW
- Combining with LaFTer further boosts performance
 - Demonstrates plug-and-play versatility
- Soft cross-entropy is weaker than equal-weight candidate sets
 - Less effective at mitigating prior bias

Mirza et al., 2023; Feng et al., 2020; Wen et al., 2021; Zhang et al., 2021b



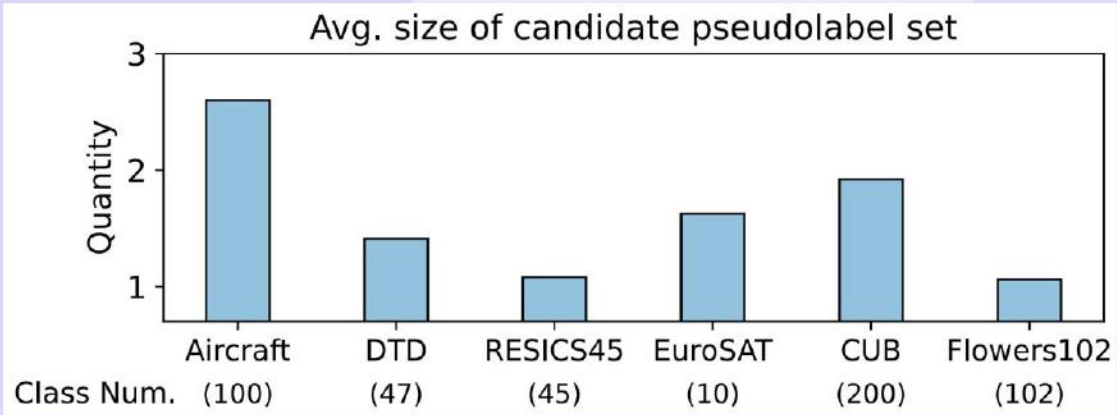
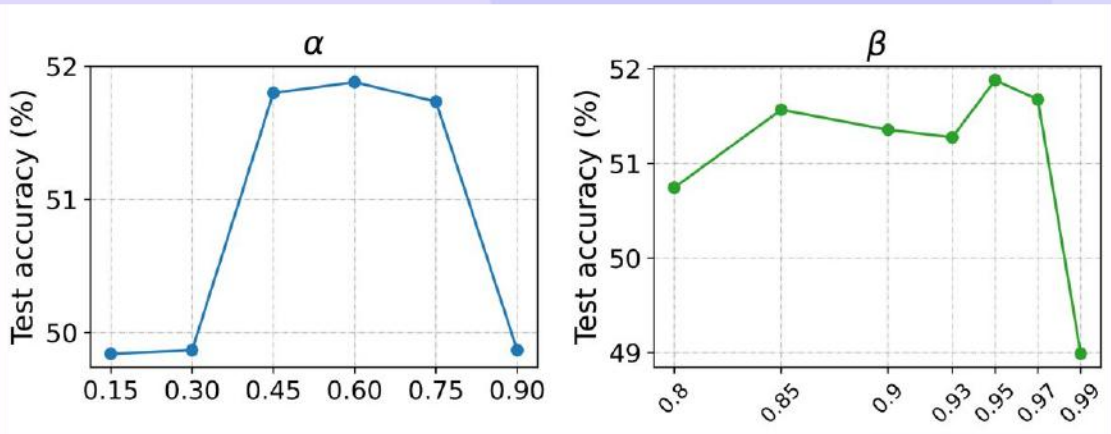
- Performance saturates as more unlabeled data is used
 - Diminishing returns beyond a point
- Small labeled sets can be more efficient on hard datasets
 - **CUB and FGVC-Aircraft** benefit from few-shot labels
- Trade off compute versus data quality when planning budgets
 - Hybrid strategies often work best

Methods	Balanced	Imbalanced $\delta=100$	Imbalanced $\delta=50$
LaFTer	74.64	65.63	66.59
CPL (w/o inter)	76.07	66.68	67.85
CPL	77.32	67.70	69.65

Methods		SSL	UL	TRZSL
DTD	Zero-shot CLIP		52.45 _{0.00}	51.61 _{0.00}
	FPL	✗	60.61 _{1.56}	52.99 _{0.43}
	CPL	✗	62.78 _{0.17}	57.23 _{0.19}
	GRIP	✓	60.91 _{0.00}	64.92 _{0.00}
	CPL	✓	69.82 _{0.32}	71.97 _{0.46}
RESISC45	Zero-shot CLIP		62.67 _{0.00}	62.13 _{0.00}
	FPL	✗	79.01 _{0.55}	70.85 _{0.66}
	CPL	✗	80.38 _{0.37}	76.01 _{0.19}
	GRIP	✓	81.53 _{0.00}	86.88 _{0.00}
	CPL	✓	87.75 _{0.29}	89.73 _{1.73}
Flowers102	Zero-shot CLIP		73.98 _{0.00}	73.05 _{0.00}
	FPL	✗	89.07 _{0.94}	77.81 _{0.30}
	CPL	✗	88.37 _{0.39}	82.98 _{0.14}
	GRIP	✓	94.21 _{0.00}	96.18 _{0.00}
	CPL	✓	96.80 _{0.63}	97.34 _{0.74}

- CPL remains superior under class imbalance
 - Inter-instance selection further helps balance
- Gains persist with larger encoders
 - Scales to stronger backbones
- Stable improvements across learning paradigms with scale
 - Maintains class balance under skewed distributions

Zhou et al., 2020



- Two key knobs control candidate size and balance
 - Moderate quantile and high selection ratio work well in practice
- Average candidate set size is close to one
 - Keeps ambiguity low for most instances
- Avoid extreme selection ratios to prevent empty sets
 - Tune per dataset with small grid search

- **Candidate pseudolabels** improve robustness and class balance
 - Stronger than hard pseudolabels across tasks
- Iterative refinement plus partial-label losses enables effective unlabeled adaptation
 - Simple to plug into existing prompt-tuning pipelines
- Versatile across prompts, learning paradigms, and encoders
 - Practical with sensible hyperparameter ranges

Thank You

Questions & Discussion